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Mapping the measurable determinants and predictive limits of endogenous A-to-I RNA editing

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Endogenous adenosine-to-inosine RNA editing reflects at least three levels of selection: whether a transcript can form a suitable duplex, whether that substrate is available to ADAR in a particular cellular state, and which adenosine is modified. The relative information carried by these levels—and the fraction inaccessible to current measurements—has not been systematically delimited. We treated prediction as an information assay and compared sequence, predicted structure, substrate organization, cellular annotations, supervision, transfer, and target-local representations across 884 human inverted-Alu substrates, with pair-disjoint development and a sealed canonical test. Sequence–structure models captured substantial within-context signal and transferred strongly among human endpoints, but less uniformly across nonhuman datasets. Inter-arm geometry explained reproducible substrate-level editing burden beyond graph summaries, whereas bulk RBP, transcript, repeat, and chromatin annotations did not resolve the latent cellular state well enough to improve site prediction. Sampling substrates before sites and aligning each target with its predicted helix partner recovered independent information; additional labels, generic capacity, frozen RNA-language embeddings, and the tested end-to-end integration gave limited or adverse returns. A frozen predictor implementing the supported principles, ADARPro, improved AUROC, AUPRC, and F1 in every held-out endpoint. Thus, current data recover intrinsic duplex suitability and target-local selectivity, while leaving transcript- and cell-resolved substrate availability as the principal measurable gap.

A-to-I RNA editing | ADAR | inverted Alu | RNA structure | machine learning

Significance

A-to-I RNA editing helps distinguish endogenous double-stranded RNA from potentially harmful signals, yet the information that selects particular substrates and adenosines remains incompletely known. We systematically tested what sequence, predicted structure, substrate organization, cellular annotations, cross-context information, and target-local context add to editing prediction. Whole-duplex and target/helix-partner information were recoverable, whereas current bulk measurements did not resolve substrate availability in individual cells or transcripts. The resulting map identifies both measurable determinants and a present predictive boundary, focusing future experiments on matched cellular state rather than on larger models alone.

Adenosine deaminases acting on RNA (ADARs) convert adenosine to inosine in double-stranded RNA (dsRNA), diversifying transcripts while preventing endogenous duplexes from activating antiviral pathways (1–5). Most detected human editing occurs in paired Alu elements, whose abundance and orientation create imperfect intramolecular duplexes (6–10). Yet potential substrates are not uniformly edited. Nearby adenosines in one predicted helix can differ sharply, and the same locus can differ among tissues (11, 12). Local sequence preference, mismatches, bulges, and helix context are established contributors (13–17), but they do not by themselves specify whether a transcript forms and retains the relevant duplex, occupies the appropriate compartment, or encounters active ADAR.

This selectivity suggests an information hierarchy. Whole-transcript sequence and folding constrain intrinsic substrate suitability; transcript- and cell-specific state governs substrate realization and access; target-local context distinguishes adenosines within an available duplex. The hierarchy is biologically plausible, but its levels have not been compared within one controlled predictive framework. Consequently, it remains unclear which aspects of endogenous editing are recoverable from existing measurements, which are obscured by the scale of those measurements, and where additional model complexity ceases to help.

Prediction is useful here as an information assay, not as a synonym for mechanism. If a candidate variable improves held-out prediction beyond

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an established sequence–structure representation, it contains reproducible information not captured by that representation. A controlled failure to improve can also be informative, but only about the tested measurement and formulation: the variable may be redundant, noisy, observed at the wrong biological scale, or genuinely uninformative. Residual error is therefore not a direct estimate of stochasticity, and predictive association is not causal identification. These distinctions allow additions, ablations, transfer tests, and saturation experiments to bound what present data support.

We used the pair-partitioned AdarEdit benchmark (18) to map this landscape across biological context, substrate organization, supervision, representation, and transfer. All development comparisons kept complete Alu pairs together and used pair-clustered uncertainty; the canonical test remained sealed until one predictor was frozen. The resulting map is asymmetric. Whole-duplex sequence and predicted structure capture broad intrinsic signal. Substrate geometry marks a reproducible burden component, but available bulk annotations do not recover the underlying cellular state. Substrate-aware learning and target/helix-partner alignment recover information lost by row-wise sampling and global pooling. Several richer representations and larger models then approach a plateau. The improved predictor is an operational synthesis of this map, not its principal scientific conclusion.

Results

A leakage-safe framework tests information at distinct biological scales. The benchmark comprises four individual GTEx tissue endpoints—artery tibial, brain/cerebellum, liver, and skeletal muscle—and Combined, a pooled endpoint rather than a fifth biological tissue. EditingIndex values at least 15% define positives and values below 1% define negatives; intermediate observations are excluded from the canonical binary task. The fixed global split contains 566 training, 139 validation, and 179 test pairs with zero overlap (Fig. 1).

Within the training and validation pairs, 5 outer folds generated out-of-fold (OOF) predictions for hypothesis testing and model selection. Feature fitting, checkpoint selection, and threshold choice were isolated from each outer fold. The internal reference was always trained and evaluated under the same folds as its candidate. Its lower absolute score reflects reduced training data and is not a replacement for published AdarEdit performance. The supplied-checkpoint canonical test remained the authoritative benchmark and was accessed only after the final model, thresholds, weights, code, and hashes had been frozen.

The comparison logic followed the proposed information hierarchy (Fig. 1A). We asked separately what whole-duplex sequence and structure capture, whether substrate-level organization contributes beyond them, whether current cellular measurements resolve residual state, which target-centered information is lost by global representations, and where additional information or capacity saturates. Evidence was classified as selected, supported but not yet actionable, limited, unsupported in the tested formulation, or untested because matched measurements were unavailable (Fig. 3A). This distinction prevents a biologically relevant variable measured at the wrong scale from being declared irrelevant.

Sequence and predicted structure capture substantial but non-universal editability. The published structure-aware graph establishes how much can be recovered before adding new information. In retained benchmark characterization runs, the compact sequence–structure baseline achieved within-context AUROC of 0.869–0.933 across the five human endpoints and 0.823–0.865 across three nonhuman datasets (Fig. 2A). These results are descriptive across different datasets, not estimates of a common effect, but they show that intrinsic RNA information carries substantial signal beyond a single species or tissue.

Transfer delineated the scope of that signal. Models transferred strongly among human endpoints, whereas transfer among nonhuman datasets was more variable and human–nonhuman transfer was lower and asymmetric (Fig. 2B and C). Thus, a shared sequence–structure grammar exists, but it is conditioned by lineage, substrate composition, assay, or cellular environment. Hand-engineered biochemical augmentation was likewise not uniformly beneficial in the retained within-context comparisons: it helped selected endpoints but reduced performance in others. More supplied features therefore did not monotonically expand the recoverable information.

These comparisons establish a strong starting point without equating discrimination with biological completeness. A sequence–structure model can rank sites while leaving substrate availability, tissue state, and unmeasured conformations unresolved. We therefore used residual gains and failures beyond this baseline to locate the remaining information rather than treating a single accuracy value as variance explained.

The RNA duplex is the effective statistical unit. The canonical row set samples a subset of the available observations, whereas the underlying training tables contain additional sites with valid high or low EditingIndex. Recovering these labels improves supervision, but uniform site sampling gives a substrate weight proportional to its number of observations. Sites within one duplex share sequence, structure, genomic context, and often editing behavior; they are not independent biological units.

A same-fold ablation separated label recovery from pair weighting in muscle (Fig. 4A; SI Appendix). Adding all high/low observations under row-wise sampling improved AUROC by +2.04 percentage points (95% pair-clustered interval +1.06 to +3.00) and F1 by +1.42 points (+0.21 to +2.57). Holding that expanded pool fixed and drawing a pair before class and site contributed a further +2.27 AUROC points (+1.25 to +3.36) and +2.85 F1 points (+1.68 to +4.01). The two corrections therefore make separable contributions.

The combined pair-first formulation improved the same-fold reference in all five endpoints (Fig. 4B). All 15 AUROC, AUPRC, and F1 intervals excluded zero, with F1 gains from +0.69 to +4.23 percentage points. The result does not imply one shared causal fate for all sites in a duplex. It establishes that the duplex, rather than an arbitrarily abundant set of its rows, is the effective statistical unit for learning from this system.

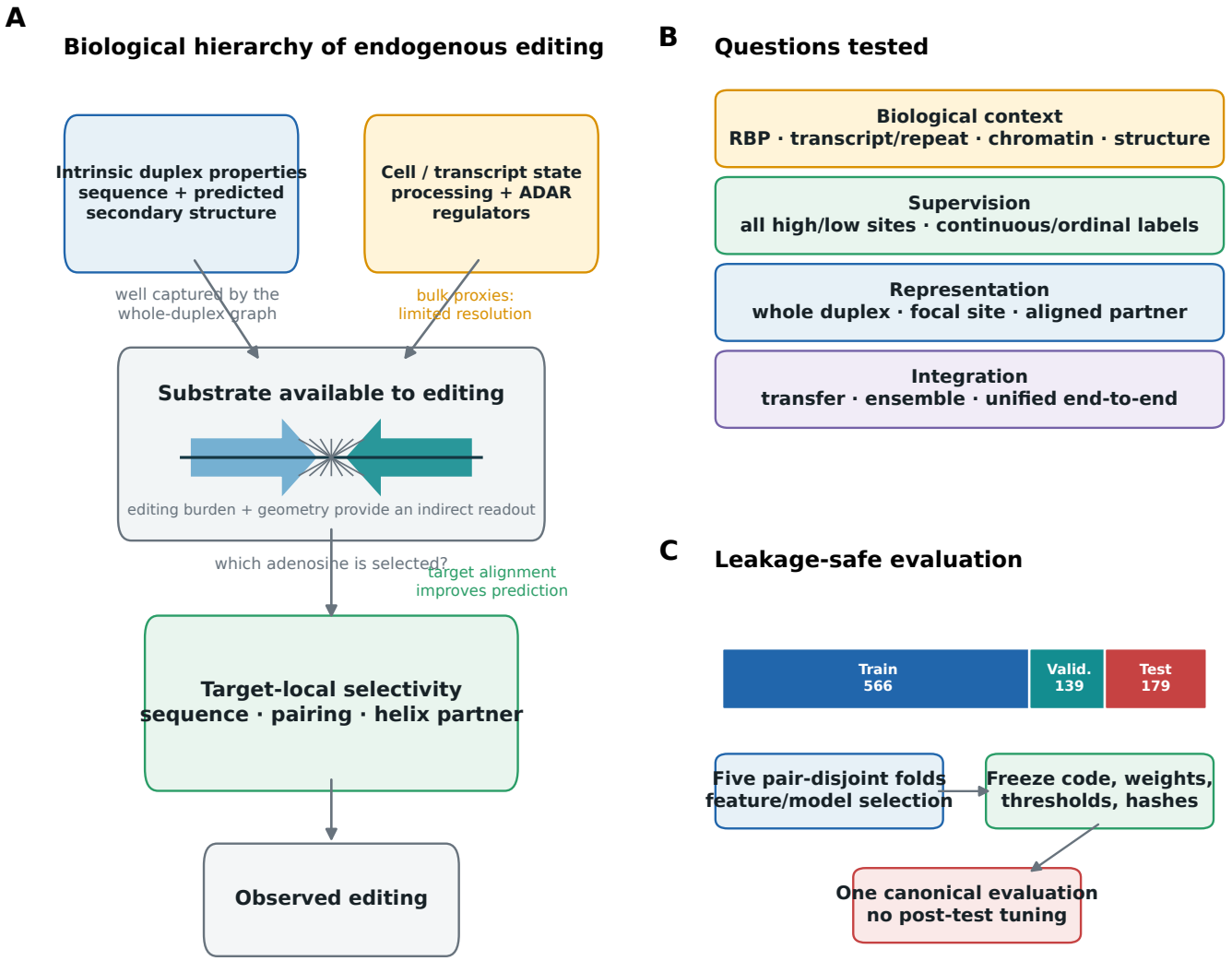


Fig. 1. An information hierarchy and leakage-safe strategy for mapping endogenous editing. (A) Intrinsic duplex properties and cell/transcript state jointly govern whether an inverted-Alu substrate is available; target/helix-partner context then contributes to selection of a particular adenosine. Pair-level burden is an indirect readout of substrate propensity, not a direct measurement of molecular engagement. (B) Controlled comparisons interrogated biological context, supervision, representation, transfer, and integration. (C) Complete pairs were partitioned globally. Development used pair-disjoint folds; the final specification was frozen before one canonical held-out evaluation. Combined is pooled and received no tissue-specific state vector.

Substrate organization marks an additional editing-propensity layer. The substrate also carries biological variation in editing burden. Genomic separation between Alu arms—information discarded when model inputs join them with a fixed separator—correlated negatively with edited-site fraction in every individual tissue (Spearman $\rho = -0.334$ to -0.482; Fig. 3B). Low-capacity models combining geometry with graph summaries predicted pair-level edited fraction with $R^2 = 0.050$–0.277 and outperformed graph summaries alone in every tissue (Fig. 3C; SI Appendix). Pair geometry thus identifies a reproducible substrate-level component of editing burden, consistent with variation in realization or engagement but not a direct measurement of either process.

Current bulk measurements do not resolve the latent cellular state. Predicting editing burden at the pair level was easier than using it to classify individual sites. A cross-fitted engagement gate added 0.0084 AUPRC but only 0.0041 F1, whose interval included zero, and increased Brier score by 0.0051. A neural engagement auxiliary likewise failed its prespecified criterion. Static geometry marks a stable propensity; it cannot reveal whether a particular transcript formed the expected duplex, remained accessible, occupied the right compartment, or encountered active ADAR.

The initial cellular-context hypothesis focused on RBP competition. We combined replicated ENCODE eCLIP occupancy, PrismNet and RBPNet predictions, literature-prioritized regulators, GTEx tissue abundance, binding-by-abundance interactions, and randomized or shuffled controls (19–22). Alu multimapping makes a missing eCLIP peak ambiguous; sequence-based predictors supplied a complementary test independent of peak absence. Neither cell-line occupancy nor tissue-average abundance contributed reproducible residual information. Cross-tissue label switches were also nearly separated by tissue identity alone, preventing four tissue means from identifying an RBP-specific transferable mechanism. These measurements are therefore unlikely to resolve

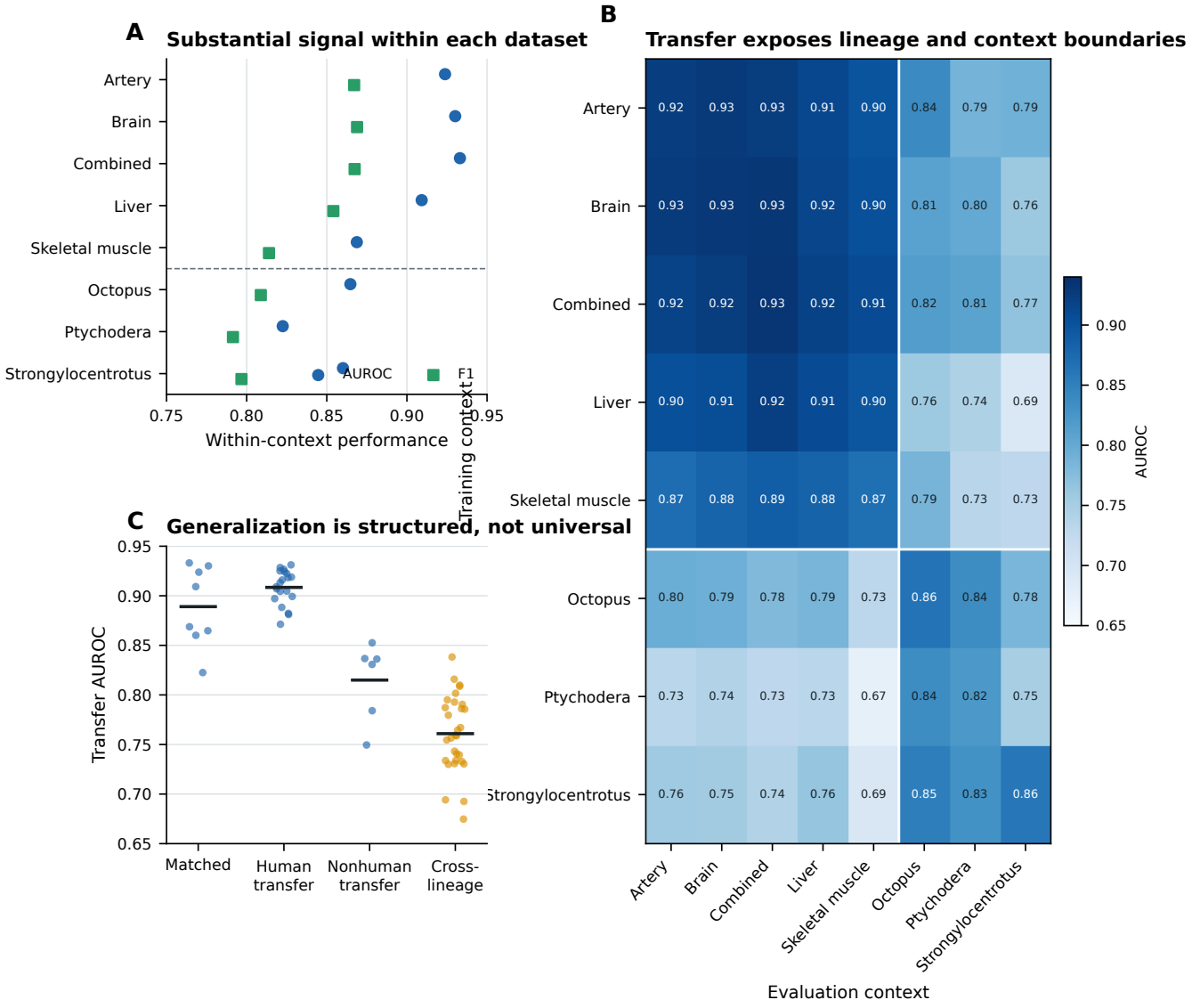


Fig. 2. Sequence–structure information is substantial but not universal. (A) Within-context AUROC and F1 for the retained compact sequence–structure baseline across five human endpoints and three nonhuman datasets. Combined is a pooled human endpoint. (B) AUROC when each training context is evaluated in every context; white lines separate human and nonhuman blocks. (C) Individual transfer values grouped by relation between training and evaluation contexts; horizontal bars are group means. These characterization runs establish predictive scope and transfer boundaries; they are not causal estimates or a meta-analysis across independent cohorts.

transcript-specific regulator exposure; matched cell- or donor-level assays are the more informative next test (23–26).

Broad transcript, repeat-subfamily, divergence, and neighborhood annotations likewise failed to improve site-level prediction beyond the graph. Exact-tissue bulk H3K36me3 overlap produced a small liver AUPRC increment without stable AUROC, F1, or pair-level benefit; no adequately matched H2A.Z resource supported a powered test. The weak bulk H3K36me3 effect points toward quantitative, sample-matched assays of cotranscriptional state rather than binary tissue-level overlap (27, 28). Together, these results sharpen the measurement problem: substrate-level propensity is detectable, but available static and tissue-average proxies do not resolve the cellular state needed to improve the strongest site classifier.

Target alignment resolves site-level selectivity. The compact AdarEdit graph captures whole-duplex sequence and structural suitability well, but averaging nucleotide embeddings can dilute the state of one target. Two intermediate experiments localized the missing context. Simple target-contact geometry improved ranking and calibration but not F1, and explicitly retaining the focal graph node improved AUROC and AUPRC more consistently than an otherwise matched duplicate-global control. Both results implicated target position before a specialized local encoder was built.

The selected local-augmented graph predictor adds a 65-position representation centered jointly on the focal adenosine and its dot-bracket-aligned helix partner. A low-capacity dilated convolution encodes nucleotide identity, pairing state, focal and partner indicators, and signed relative position; its output is fused with the whole-duplex graph embedding before

A

What adds to the sequence-structure graph?

Tested formulation	Evidence	Main finding
LEARNING & SUPERVISION		
Pair-first expanded supervision	Selected	More labels; correct pair weight
Continuous + ordinal auxiliary labels	Limited	Small ranking gain
Shared four-tissue training	Limited	Transfer below preset criteria
TARGET REPRESENTATION		
Aligned target + helix-partner view	Selected	Passes shifted-window control
Focal node / contact geometry	Limited	Localizes target effect
Frozen RNA-language embeddings	Not supported	F1-only; no ranking gain
Substrate-wide all-A decoder	Not supported	Stopped for futility
Larger / deeper graph networks	Not supported	No consistent benefit
Graph Transformer / GATv2 expansion	Not pursued	Simpler validated model sufficient
SUBSTRATE & CELLULAR CONTEXT		
Pair geometry / editing burden	Supported*	Substrate propensity detectable
Hierarchical engagement gate	Limited	Ranking gain; unstable F1
Bulk H3K36me3 overlap	Limited	Liver AUPRC only
RBP binding × bulk expression	Not supported	No residual gain
Transcript / repeat annotation panel	Not supported	No site-level gain
Matched donor/cell multimodality	Deferred	Matched data unavailable
Matched in vivo structure / isoforms	Deferred	Matched data unavailable
INTEGRATION		
Fixed whole-duplex + local-augmented logits	Selected	Complementary predictions
Unified end-to-end architecture	Not supported	Below frozen ensemble

See Table S2 for comparators, effect sizes, and intervals.
*Reproducible observation not yet useful for site prediction; deferred hypotheses were not tested.

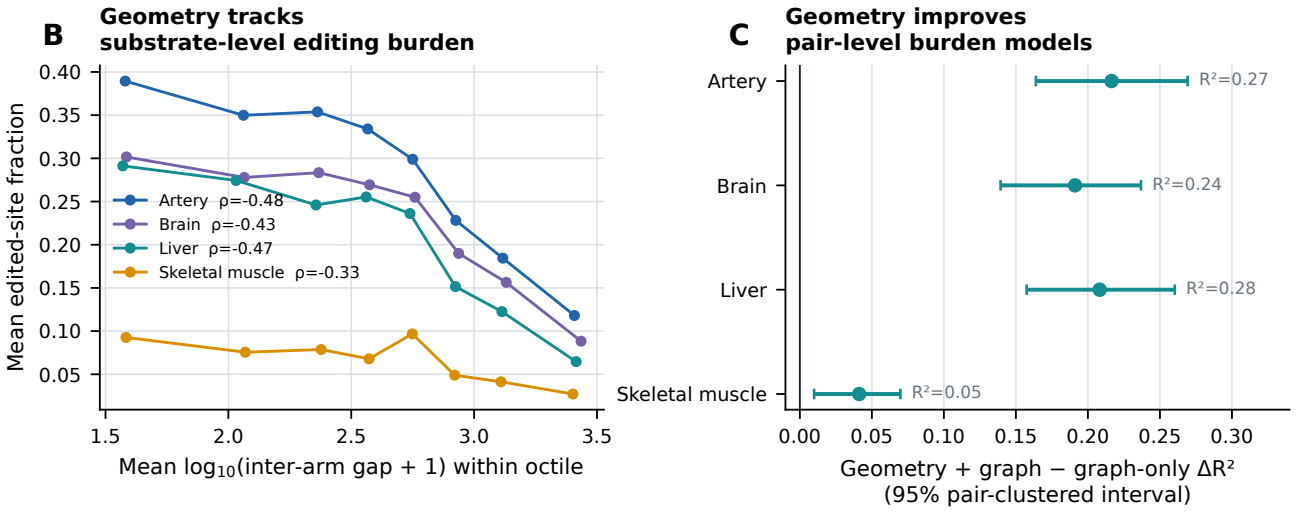


Fig. 3. A substrate-level signal remains unresolved by current context measurements. (A) Tested information sources grouped by scale and outcome; SI Appendix, Table S2 gives each comparator, effect size, and interval. "Not supported" applies only to the tested formulation, deferred measurements were unavailable, and superseded branches were not interpreted as negative. (B) Mean edited-site fraction across octiles of inter-arm gap in four tissues; ρ is calculated over individual development pairs. (C) Increment in pair-level R^2 from adding genomic geometry to graph summaries. Bars are 95% pair-clustered bootstrap intervals; labels give absolute R^2 for the combined model.

classification. In muscle OOF data it reached AUROC/AUPRC/F1 of 0.9045/0.8936/0.8337, compared with 0.8885/0.8730/0.8208 for the pair-first whole-duplex-only predictor (Fig. 4C).

The decisive control retained the same graph-plus-convolution architecture and optimization but shifted the local center by 96 nt. Performance fell to 0.7122/0.6943/0.7028. Aligned-minus-shifted differences were 0.1926 AUROC, 0.1993 AUPRC, and 0.1311 F1, with entirely positive pair-clustered intervals. This matched-capacity shift sharply reduced performance, localizing the gain to target alignment rather than added convolutional capacity. Liver replication and the secondary natural-prevalence analysis showed the same pattern.

Whole-duplex and target-local information remain complementary. The local-augmented predictor did not make an independently optimized whole-duplex view redundant. Their errors remained sufficiently distinct that a fixed mean of logits improved the whole-duplex-only predictor by 0.0256/0.0308/0.0139 AUROC/AUPRC/F1 in muscle

and 0.0182/0.0214/0.0179 in liver; all six pair-clustered intervals excluded zero (Fig. 4D).

The final predictive specification consequently used two independently trained pair-first components: a whole-duplex-only graph predictor and a second graph predictor internally augmented with the aligned target/helix-partner convolution. Their logits are combined by a fixed 0.5/0.5 mean; the weights were not searched (Fig. 4E). Fixed integration preserved complementary biological views without fitting a flexible stacker to only hundreds of independent substrates.

Additional information and model complexity approach a plateau. Intermediate EditingIndex values contain graded information, but most useful thresholded content was already recovered by pair-first binary training. A binary/ordinal/continuous multitask head increased AUROC by 0.0081 (95% pair-clustered interval 0.0010–0.0151) while changing F1 by only 0.0008; its integration was adverse under natural prevalence. Shared four-tissue training likewise showed transferable rules, with mean changes across the four individual tissue endpoints of 0.0087 AUROC, 0.0114 AUPRC, and 0.0075 F1, but fell below its predeclared selection criteria and worsened mean Brier score by 0.0043. Graded labels and cross-tissue sharing therefore refine ranking, but less than correcting supervision and target representation.

Other increases in capacity were unproductive. A substrate-wide decoder trained all candidate adenosines jointly changed AUROC by -0.0664 and F1 by -0.0599 relative to the pair-first model and met its prespecified two-fold futility boundary. Frozen RiNALMo target and substrate embeddings, including a larger model, added little beyond the task-specific structure graph (29). Larger or deeper graph networks likewise failed stronger-model controls. More parameters did not compensate for misaligned supervision or representation. The sharper pretraining question—benchmark-safe, inverted-Alu-specific learning—remained deferred because a leakage-safe corpus could not be guaranteed.

Before test access, we preregistered one connected architecture to ask whether the successful components would interact more effectively during joint optimization. It combined a three-layer graph encoder, focal-node state, aligned local encoder, pairwise representation interactions, and a training-only substrate head. Its 65,698 trainable parameters were fewer than the 80,138 parameters across the two ensemble components. On the 5,967-row, 585-pair muscle development set, the unified model reached 0.9072/0.8922/0.8317 AUROC/AUPRC/F1 versus 0.9142/0.9040/0.8347 for the frozen ensemble. AUROC and AUPRC were worse in every fold; their pair-clustered intervals were wholly adverse, and Brier increased by 0.0205 (0.0135–0.0276). Under the preregistered decision rule, repair and replication stages were not opened. Across these tests, improvements saturated when added inputs did not better match the biological scale of the unresolved information. This is an empirical boundary for the tested measurements, representations, and sample size, not a universal upper bound for A-to-I editing prediction.

A frozen predictor validates the information map on the untouched benchmark. After development selection, 10 component fits were trained on published training pairs. Published validation data alone selected component checkpoints and epochs and one ensemble threshold per endpoint. The exact code revision, configurations, weights, and hashes were bound into a final manifest. A single authorization then exposed the canonical test for one evaluation; no model or claim-defining analysis was changed afterward.

The resulting predictor, ADARPro, improved AUROC, AUPRC, and F1 in all five reported endpoints (Fig. 5A). Mean gains over reproduced supplied-checkpoint AdarEdit were 0.0277, 0.0261, and 0.0253. Because Combined overlaps the individual tissue endpoints, this mean is descriptive rather than a five-tissue meta-analysis. F1 changed from 0.8671 to 0.8869 in artery, 0.8689 to 0.8902 in brain, 0.8541 to 0.8812 in liver, 0.8139 to 0.8541 in skeletal muscle, and 0.8674 to 0.8852 in Combined. Every endpoint had a positive 95% pair-clustered bootstrap interval for all three metrics, and Brier score decreased throughout (Fig. 5C). The canonical gain confirms that the substrate-aware and target-aligned principles selected during development carry independent held-out information.

The gains were not purchased by indiscriminate suppression of positive calls. At frozen validation-selected thresholds, ADARPro corrected 32.0–50.6% of baseline false positives while damaging 1.7–5.9% of baseline true positives (Fig. 5D). In a post-test descriptive analysis that rethresholded ADARPro to match AdarEdit recall exactly, the point estimates were 126, 119, 158, 68, and 148 fewer false positives across artery, brain, liver, skeletal muscle, and Combined; every pair-clustered interval excluded zero (SI Appendix, Table S9).

Discussion

This study maps predictive information rather than claiming a complete causal decomposition of RNA-editing biology. The map has three levels. Whole-duplex sequence and predicted structure encode a broadly transferable component of intrinsic suitability. Pair organization marks a substrate-level propensity that is reproducible but only partly actionable. Within a plausible substrate, target/helix-partner context adds information that whole-duplex pooling obscures. A fourth conclusion concerns inference itself: sites within one duplex are statistically dependent, and effective substrate weighting matters independently of recovering more labeled observations.

The positive and negative results together define the map. Strong within-context performance and human transfer show that intrinsic RNA information is substantial; variable nonhuman and cross-lineage transfer show that it is not universal. Pair geometry adds information about burden but not a stable site-level decision rule. Cell-line RBP occupancy, tissue-median abundance, broad transcript and repeat annotations, and bulk chromatin do not recover the missing state. Continuous labels and shared tissue learning add small ranking signals, whereas generic frozen RNA embeddings, an all-adenosine decoder, greater graph capacity, and the tested unified architecture do not close the gap. These are constraints on current formulations, not evidence that the corresponding biology is absent.

The residual errors consequently have several nonexclusive interpretations. They may reflect donor- or cell-

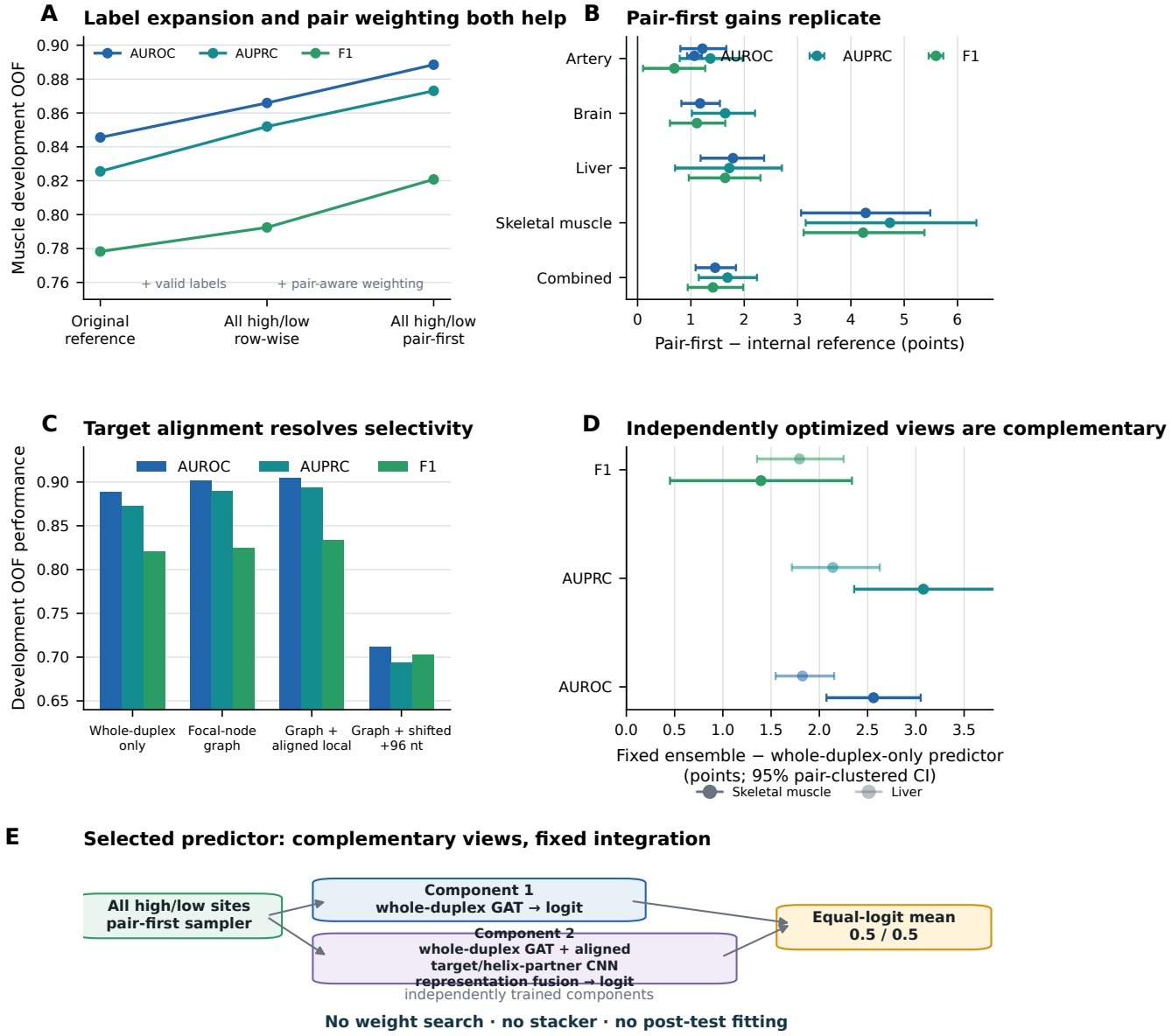


Fig. 4. Substrate-aware learning and target alignment recover complementary information. (A) Same-fold ablations separate recovery of additional valid high/low labels from pair-first weighting. (B) Pair-first expanded supervision improves the internal reference across all five endpoints; points and bars are pair-clustered medians and 95% intervals. (C) Whole-duplex pooling, focal-state retention, and aligned-local fusion form an evidence ladder; an otherwise identical local model shifted by 96 nt fails. (D) A fixed equal-logit ensemble improves its whole-duplex-only component in muscle and liver. (E) The operational predictor combines independently trained whole-duplex-only and graph-plus-local models without a searched weight, stacker, or post-test parameter. All panels show development evidence.

specific ADAR abundance and isoform state, interferon response, transcript processing and residence time, regulator occupancy, or in vivo conformations absent from predicted minimum-energy structure. They may also include measurement error, imperfect representation, finite training data, and biological stochasticity. The present experiments cannot apportion those sources. What they establish is narrower and useful: once whole-duplex and aligned target information are represented and substrate dependence is respected, generic feature and capacity additions show diminishing returns, while measurements matched to individual transcripts and cellular states remain absent.

This framing also separates statistical correction from biological inference. Pair-first sampling corrects unequal influence from site-rich duplexes; it does not imply that every adenosine in a substrate shares one causal fate. Pair-level burden, conversely, supports a stable substrate propensity even when it fails to improve the leading classifier. The success of target/partner alignment localizes predictive information near the edited adenosine and its paired context, but does not identify a biochemical mechanism. The worse preregistered unified model further shows that a biologically connected architecture is not automatically more informative at this effective sample size.

Several limitations bound generality. The primary benchmark comprises graph-buildable human inverted-Alu substrates and tissue summaries, not all editing substrates or donor-resolved measurements. The nonhuman

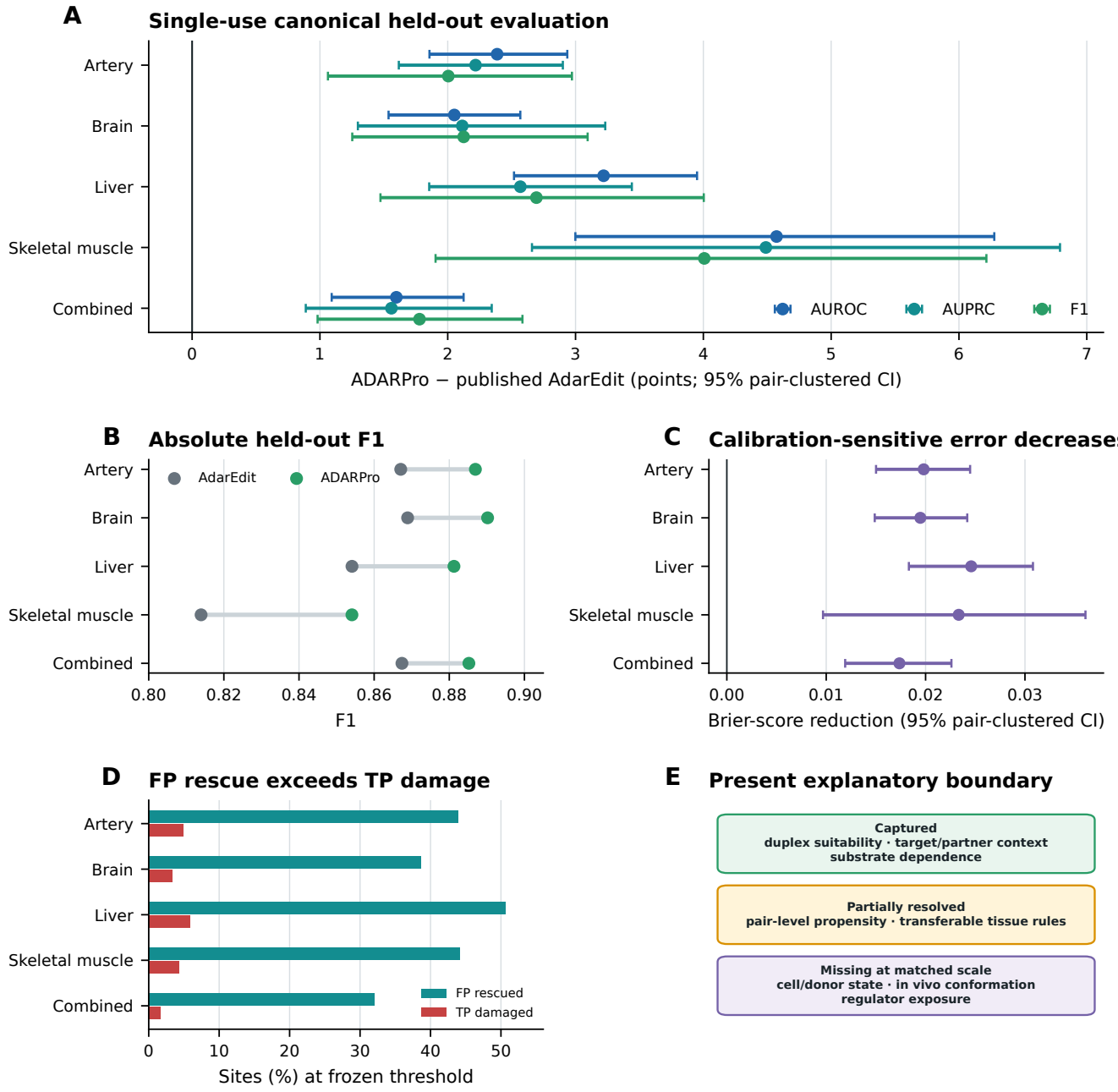


Fig. 5. Frozen validation confirms the map and locates the present boundary. (A) ADARPro minus reproduced supplied-checkpoint AdarEdit for AUROC, AUPRC, and F1; bars are 95% pair-clustered bootstrap intervals. (B) Absolute F1. (C) Reduction in Brier score, a calibration-sensitive error measure. (D) Baseline false positives corrected and baseline true positives damaged at frozen validation-selected thresholds. (E) Synthesis of information captured, partially resolved, and missing at a matched biological scale. Combined is pooled. Quantitative panels were generated after the final manifest was frozen and the one-use authorization was consumed.

matrices broaden the scope assessment but differ in lineage, substrate class, and data generation, so they do not isolate the cause of transfer loss. Predicted secondary structure is a surrogate for conformations available to ADAR in vivo. The canonical binary task excludes intermediate observations and is balanced; natural-prevalence analyses are secondary. eCLIP coverage is cell-line-specific and compromised by repeat mapping. Finally, the canonical partition is stringent but originates from the same benchmark construction; an independent cohort is needed to test transportability.

ADARPro is therefore best understood as a validated instrument produced by the information map. Its held-out improvement demonstrates that substrate-aware supervision and target-centered representation contain recoverable signal beyond the published predictor, but the larger conclusion is the boundary it helps expose. The next decisive study should measure editing, transcript abundance and processing, splicing and intron retention, ADAR1 isoform and interferon state, ADAR activity and localization, regulator occupancy, quantitative chromatin, and in vivo RNA conformation in the same donors or cells and on the same substrates. Such matched measurements could test whether cellular state governs realization of an intrinsically suitable duplex while local RNA context governs selection of particular adenosines.

Materials and Methods

Benchmark and labels. We used the AdarEdit v2 reproducibility release at upstream revision `5c8b57d70cdc2eb1c5e168431013275bcc9ed0c5`. The benchmark contains 884 graph-buildable inverted-Alu pairs. Human labels were edited for EditingIndex $\geq 15\%$, nonedited for EditingIndex $< 1\%$, and intermediate otherwise. The canonical task excludes intermediate observations. Pair identities and the fixed 566/139/179 train/validation/test assignment were verified before analysis. Combined was treated only as its published pooled endpoint.

Development protocol. All discovery used the original train plus validation pairs divided into 5 outer folds by pair ID. Learned preprocessing, feature selection, checkpoint choice, and threshold choice were fitted without the outer fold. Predictions were concatenated only after every row was scored by a model that had not trained on its pair. The test paths were protected by a stateful quarantine. Development OOF metrics estimate same-fold incremental effects; supplied-checkpoint test metrics remain the authoritative reference.

Models. The first component retained the compact AdarEdit graph attention encoder and trained from the expanded high/low site pool using uniform pair-first and equal-class sampling. The second, independently trained component retained the same graph encoder and added a 65-position target/helix-partner-aligned representation encoded by two dilated convolutional layers; local and graph embeddings were fused before its prediction head. The selected probability was $\sigma[(\mathrm{logit}\, p_{\mathrm{whole}} + \mathrm{logit}\, p_{\mathrm{augmented}})/2]$; weights were fixed at 0.5/0.5 rather than searched. The shifted-window control preserved the graph-plus-convolution architecture and optimization while moving the local center 96 nt. Full feature definitions, architectures, ablations, and biological-context protocols are in SI Appendix.

Statistics and final evaluation. Metrics were AUROC, AUPRC, F1, precision, recall, specificity, balanced accuracy, MCC, and Brier score. Thresholds maximized validation F1 within each frozen training run. Uncertainty used a pair-clustered bootstrap with 2,000 replicates that resampled Alu pairs and retained all their sites. For the post-test descriptive matched-recall analysis, the candidate threshold was selected on final predictions solely to equal supplied-checkpoint baseline recall; uncertainty again resampled pairs. Before final evaluation, code, inputs, weights, epochs, thresholds, and checkpoint SHA-256 values were frozen. Test authorization was consumed once, and no post-test model selection or tuning occurred. Natural-prevalence results use all eligible high/low observations from development pairs under unchanged pair assignments and are secondary to the canonical balanced task.

Data, Materials, and Software Availability. Code, experiment configurations, registries, source tables, figure scripts, retained derived artifacts, frozen checkpoints, and provenance are available at https://github.com/Scientific-Computing-Lab/ADARPro. Third-party downloads and deterministic matrices omitted from GitHub are listed with source URLs, roles, checksums, and regeneration instructions in the data manifest. The AdarEdit benchmark is available from https://github.com/Scientific-Computing-Lab/AdarEdit. No new human participants were recruited and no new primary sequencing data were generated.

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